# Ahmad Akil

Sylhet, BD | ferozbinakil@gmail.com | +8801751578424

Linkedin.com/in/akil67
Github.com/ahmadakil67
Sites.google.com/view/ahmad-akil-portfolio
Kaggle.com/ahmadakil


## EDUCATION

**North East University Bangladesh, Sylhet**

BSc (Eng) in Computer Science & Engineering — Jun '20 – Sep '24

- Cumulative GPA 3.38/4.00
- Relevant Coursework: Data Structure and Algorithm, OOP, Machine Learning, Deep Learning, Database Management System, Computer Networking, Web Engineering, Software Engineering, Communication Engineering, Artificial Intelligence

## UNDERGRAD THESIS

**Manipulated Image Detection: Pixel-Level and Patch-Wise Analysis** Code | Report — July '23 – July '24

- Developed LSTM-based pixel-level analysis for detecting subtle image manipulations.
- Implemented ConvNet architecture for patch-wise analysis of large alterations.
- Achieved 92% accuracy in manipulated image detection using the LSTM-ConvNet hybrid model.
- Leveraged statistical properties of images to enhance detection accuracy.

## SKILLS

**Machine Learning:** Linear Regression, Logistic Regression, Random Forest, KNN, Naive Bayes, SVM, Decision Tree, K-Means, Hierarchical Clustering, XGBoost, PCA

**Deep Learning:** ANN, CNN, RNN, GRU, LSTM, YOLO, NLP, Transfer Learning, Object Detection, Transformer

**Programming Language:** Python, C++, C, Java, JavaScript, SQL, HTML, CSS

**Python Packages and Frameworks:** OpenCV, TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, Matplotlib, Pytorch

**Mathematics for ML/DL:** Algebra, Probability, Statistics, Calculus, Matrices

**Version Control:** Git & GitHub

## PROJECTS

**Machine Learning Project: Kidney Disease Classification** | *Python, Computer Vision* — May '24

- Utilizing VGG16 architecture to develop a model for classifying kidney diseases into four categories: Cyst, Normal, Stone, and Tumor.
- Sourcing the dataset from Kaggle, which includes labeled medical images for training and testing.
- Project is currently in progress, focusing on optimizing model accuracy and robustness**.**

**Predict Closed Question on Stack Overflow** | *Python, NLP, TensorFlow, Kaggle notebook* — Nov '23

- Predict Off-topic, Not Constructive, Too Localized, and Not Real Questions on Stack Overflow.
- Dataset is collected from **Kaggle**.
- The **BERT** model achieved an impressive **84%** accuracy, surpassing SOTA benchmarks. The **MLP** model showed a solid **64%** accuracy, and **Naive Bayes** exceeded expectations with an accuracy of over **24%.**

**Stock Sentiment Analysis using News Headlines** | *Python, NLP, Pandas, Scikit-Learn, Kaggle* Dec '22

- Predict whether today's stock price will increase or decrease.
- Applied machine learning models are Neural Network, XGBoost, Decision Tree, and Support Vector Machine.
- The Neural Network surpassed the state-of-the-art performance, achieving an accuracy of **98%**.

**Desktop Chatting Application** | *Java, Java Swing, Socket* Feb '21

- A client-server chat application built using Java Socket programming where two clients can connect to a server and chat among themselves.

## ACHIEVEMENTS

- Ranked 175 out of 225 teams at the ICPC Asia Dhaka Regional Contest-2023
- Participated in the 13th National Undergraduate **Mathematics Olympiad** 2022.
- Solved **500+** problems with Python in various platforms (e.g., Codeforces, Leetcode).

## CERTIFICATIONS

- Machine Learning Specialization | *DeepLearning.AI*
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- Machine Learning A to Z | *Udemy*